Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Variation of Parameters method

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Variation of Parameters method

1 Variation of Parameters method

Study focus: Variation of Parameters method. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000226 · EV0000227

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This method is used to find the particular integral of a linear differential equation whose complementary function is already known.

Key details

- So, this method is used to find only the particular integral of the differential equation.
- To find the particular integral of the differential equation, the complementary function must be known.
- Though this method is very general, we will apply this method.
- To solve a second order linear differential equation with constant coefficient.
- P and Q are constants and capital X is a function of X.
- See, I have considered a simple second order non-homogeneous linear differential equation with constant coefficients.
- I know to find the complementary function of this differential.
- And we know that this c1 by 1 plus c2 y2 represents the complementary function of the differential equation only when y1 and y2 are two.
- Two linearly independent solutions of the associated homogeneous linear differential equation.
- When do we say that y 1 and by 2 are two linearly independent?
- Now, because this c1 y1 plus c2 y2, we have taken it as the complementary function, y1 and y2 have to be the linearly independent functions.
- Y1 by 2 and its derivative y 1 dash by 2 dash.
- That is w is equal to y1 into y2 dash minus y2 into y1 dash.
- Y2 is not equal to 0 if y1 and y2 are two linearly independent functions.
- So because here y1 and y2 are linearly independent, so w is not equal to 0.

- What is capital X?
- What are y1 and y2?
- Two linearly independent solutions of the corresponding.
- Homogeneous differential equation.
- I need to evaluate these two.
- To find the particular integral by this method.
- But when I am writing the expression for u and b, I am not writing the constant of integration because the constant of integration gets observed with the constants of the complementary function when I'm writing the general solution of the differential equation.
- So this method basically involves finding out are evaluating the integrals u and b and finding out this Ronski and W.

Example or application from the lecture

- Let us introduce another term called Ronskeian.
- P i is equal to some u of u a function of x, u of x into y1 plus v of x into y2, where u and v are given by u is equal to minus integral y2 into x by w d and u v is equal to the equal to equal to integral y1 into x by w dx.
- That is the right hand side function of the given differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 05:00–09:59

non-homogeneous linear differential equation with constant coefficient.

Key details

- This method can also be used to find the particular integral of.
- Our differential equation with variable coefficient.
- If we know the general solution of the associated homogeneous equation, in the sense this method can also be used to find the particular integral of LTE with variable coefficient provided.
- This method can be applied to solve a differential equation of any order that is and th order linear differential equation.
- But we are learning in this session m to solve only LTE of order 2.
- When we are writing, when we are finding the using this method to find the solution of the differential equation, we should note down this.
- If it is not equal to one any other constant number, divide by the coefficient and make the coefficient of d2 y by dx to one.
- What are the demerits of this method?
- See what is the main demerit of this method?
- This method is used to find only the particular integral of the differential equation and.
- To find the particular integral also we need to know the complementary function of the differential equation.
- So the complementary function is absolutely required to apply this method.

- And this method involves evaluation of some integrals, which are sometimes very difficult to evaluate or solve.
- So these are the main two demerits of this method.
- Linear differential equation with constant coefficient and with variable coefficient, but we are applying this method to find the solution of only second order LBE.
- Find the particular integral by the method of variation of parameters.
- It is capital D square minus two d minus three of y is equal to x into e to the power minus x.
- D by dz or it is not d by dz.
- We have we are discussing a different method here, variation of parameter method.
- So it is because the independent variable here is x, so d is d by dx.
- We know to solve this differential equation.
- We know to find the complementary function and also the particular integral of this differential equation which we learnt in the previous session.
- How do you find the particular integral of this equation?
- This is type 4.
- It is e to the power ax into v type where v is a function of x.
- So using that method I can find the particular integral.
- But here the question is to find the particular integral using the method of variation of parameters.
- To find the particular integral of this differential equation, by using method of variation of parameter, I need to write the complementary function of the differential equation.
- If the question was to find the particular integral,.
- To find the particular integral by any method, I need not have to find the complementary function because it is by the method of variation of parameters I need to first find out the complementary function of the differential equation.
- So this method involves three steps.
- The first step being writing the complementary function and finding out by one and by two.
- The second step is to find the RAN scale and the third step is to find u and b.
- function is c 1 e to the power 3x plus c2 to the power minus x.
- C e to the power 3x and e to the power minus x are 2.

Example or application from the lecture

- We know the complementary function of the differential equation, and this method we illustrated by taking the example of second order differential equation.
- The coefficient of d2y by dx2 in the given differential equation has to be 1.
- Now let us apply this method to solve.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

Core explanation · 10:00–14:59

linearly independent functions.

Key details

- Derivatives of it three into e to the power three x minus e to the power minus x is the second row.
- two x minus three e to the five.
- e to the power 3x into minus e to the power minus x minus half 3 e to the power 3x into e to the power minus x which simplifies to minus 4 into e to the power 2x.
- The particular next the last step is to find the particular integral.
- Minus what is y 2? e to the power minus x.
- By W which we found out it is minus 4 into E to the power 2x.
- Here you have e to the power minus 2s, which simplifies to 1 by 4 e to the power minus 4s into xpx.
- 1 by 4 into C.
- I have taken x as the first function and I'm integrating e to the power minus 4x.
- integral of integration of this is this minus e to the power minus four x by four into differentiation of this x.
- with respect to x is 1 dx.
- This is just one integral evaluation to find u.
- function b to find the particular integral of the differential equation.
- Integration y 1 into x by w dx.
- What is x? x into e to the power minus x.
- W the wrong scale which we found out it is minus 4 into e to the power 2x.
- E to the power 2x gets canceled with e to the equation.
- by 2x it is minus 1 by 4 into integration of x dx.
- It is x square by 2.
- See I told you when you are calculating u and v no need to write the constant of integration.
- And v is equal to x square by 8.
- See this is the particular solution.
- Particular solution should not contain any constant.
- If the question was to find the complete solution, then you should write the complete solution.
- Y is equal to C plus PI.
- We learn this method to find the particular integral of a second order differential equation with constant coefficients, but this method can be applied to solve linear differential equation of any order with constant coefficient or with variable coefficients.
- So we will solve more problems on this method in the next session.

Example or application from the lecture

- The Ronskiana by one by two is given by determinant e to the power three x e to the power minus x that is the first row.
- The particular integral is given by u into y 1 plus v into y2, where u is e minus integral y2 into x by w dx and b is.
- And v is given by integral y 1 into x by wtx.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Variation of Parameters method transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00054	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00056	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice